

Supervised learning algo ✓

Another reason we wanted to model $p(t)$

for $t \in T_G$ is that could use p to find most likely t for given sentence

given q ^{PCFG+} and target sentence s

Want: $\arg \max_{t \in T_G(s)} p(t)$ ✓

Other desiderata

$p(s)$?

$$= \sum_{t \in T_G(s)} p(t)$$

$p(s) > 0 \iff s$ is grammatical

$|T_G(s)| \leftarrow$ also interesting

Chomsky Normal Form A

CFG G is in C.N.F. if

every rule $(X \rightarrow \beta) \in R$, either

① $X \rightarrow Y_1 Y_2$, $Y_1, Y_2 \in N$

② $X \rightarrow Y$ $Y \in \Sigma$

Prop¹ For any CFG G , $\exists$ CFG in C.N.F. G' that is the same grammar

s is grammatical in $G \iff$ grammatical in G'

Prop² For any PCFG $G \exists$ PCFG in C.N.F. G'

st $\forall s \quad P_G(s) = P_{G'}(s)$

isomorphism

...

Moreso $\mathcal{T}_G(s) \leftrightarrow \mathcal{T}_{G'}(s)$ have a bijection that preserves probabilities

$$t \leftrightarrow t'$$

$$p_G(t) = p_{G'}(t')$$

you'll prove it c

Finding max tree in C.N.F. PCFG

$$\arg \max_{t \in \mathcal{T}_G(s)} p(t)$$

$$s = x_1, \dots, x_n \quad x_i \in \Sigma$$

Brute Force?

PCFG G

$$X \rightarrow Y_1 Y_2$$

$$X \rightarrow x \in \Sigma$$

Most brute force:

obscure need exactly n non-terminals

to get length n yield

$\therefore$ need to apply $(n-1)$ rules

enumerate $|R|^{n-1}$ lists of $(n-1)$ rules,

for any that can yield $x_1, \dots, x_n$ is valid and

take $p(t)$, keep max $O(|R|^{n-1})$



DP to beat brute force

CKY algorithm (CYK also)

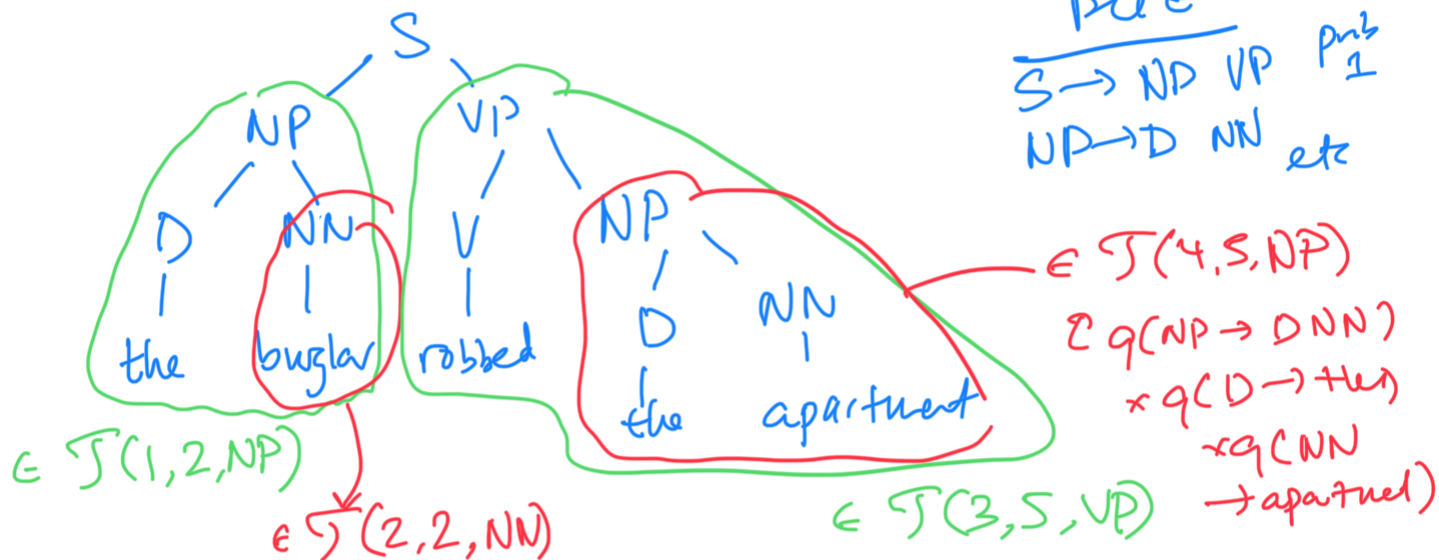
$$s = x_1 \dots x_n$$

$$x_i \in \Sigma$$

Define $\mathcal{T}(i, j, X)$ ($1 \leq i \leq j \leq n$)

= set of all "partial" parse trees t over just the

Words $x_i \dots x_j$ with X as the root



DP Table $\pi(i, j, X) = \max_{t \in T(i, j, X)} p(t)$ $\leftarrow$

$p(t) = \prod q(\text{rules used in deriving the partial tree})$

max prob of any parse tree over words (with yield)
 $x_i, \dots, x_j$ where X is the root / start symbol (need not be S)

FP DP 1 Solve final problem

$\arg \max_{t \in T(1, n, S)} p(t) = \pi(1, n, S)$

FP DP 2 recursive constructability

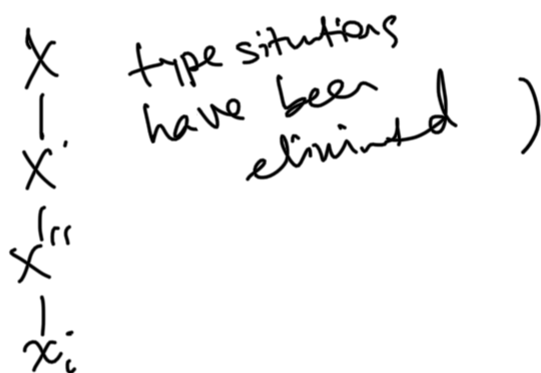
going to build by increasing $(j-i)$

Base case, $|j-i| = 0$

$\forall i=1, \dots, n$

$$\pi(i, i, X) = \begin{matrix} \text{max prob} \\ \text{of subtree} \\ X \\ \downarrow \\ x_i \end{matrix} = \begin{cases} q(X \rightarrow x_i) & \text{if } X \rightarrow x_i \in R \\ 0 & \text{otherwise} \end{cases}$$

(we use that in C.N.F. CFG, any terminal $x_i \in \Sigma$ must come 'immediately' from non-terminal X at root of subtree)



Recursion, suppose all smaller gaps $j-i \in \{0, \dots, g-1\}$ and all $X \in N$, $\pi(i, j, X)$ filled in

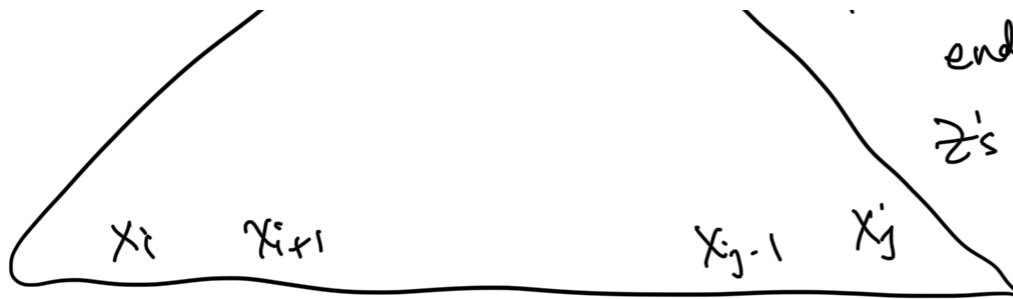
now we have $j-i = g$

$$\pi(i, j, X) = \max_{\substack{X \rightarrow YZ \in R \\ s \in \{i, \dots, j-1\}}} q(X \rightarrow YZ) \times \pi(i, s, Y) \times \pi(s+1, j, Z)$$

pf by picture For $j-i > 1$ maximizing partial tree w/ root X over $x_i, \dots, x_j$ must use some $X \rightarrow YZ$ at the top



now Y, Z roots of their own separate subtrees Y 's span has to start at x_i



ends at some x_j
 Z 's span starts at x_{i+1} and ends at x_j

? is between i (if $? = i$, then $Y \rightarrow x_i$ is the next rule to get the whole left subtree)

and $j-1$ (can't be j , otherwise $Z \rightarrow$ nothing??, not allowed)

if $? = j-1$, Z spans only x_j so $Z \rightarrow x_j$ is the expansion rule of Z

But you don't know which rule $X \rightarrow YZ$ is used in the most likely derivation. Nor do you know where $?$ is for that rule

so take max over $X \rightarrow YZ$ and midpoint of the ranges of Y vs Z $s'' \in \{i, \dots, j-1\}$
 $\square$

CKY algo pseudocode

// Base case

for $i \in \{1, \dots, n\}$, for $X \in N$

if $X \rightarrow x_i \in R$

$$\pi(i, i, X) = q(X \rightarrow x_i)$$

else $\pi(i, i, X) = 0$

// Recursion

```

// gap size
for g = 1, ..., (n-1)
  for i = 1, ..., (n-g) // looping over start index,
                        // can't be < gap away
                        // from n
    j = i + g
    for X ∈ N
      π(i, j, X) = max_{X → YZ ∈ R, s ∈ {i, ..., (j-1)}}
                    q(X → YZ) × π(i, s, Y)
                    × π(s+1, j, Z)
  // also a loop over |R| × (j-i)

return π(1, n, S)

```

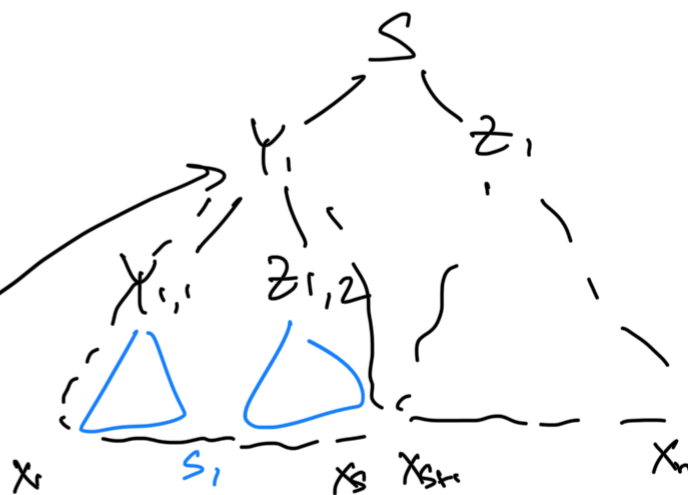
What if we wanted tree itself not just prob of it?
 Backpointer, like Viterbi

For every "max tree over $x_i, \dots, x_j$ w/ root X "
 also SAVE rule $X \rightarrow YZ$ + s that witnessed/
 gave that max tree

read out: $\pi(1, n, S)$

best rule
 $S \rightarrow Y_1 Z_1$
 and witness s

$\pi(1, s, Y_1)$
 $Y_1 \rightarrow Y_{1,1} Z_{1,2}$
 s_1



Time complexity of CKY (with or w/o backpointers)

$$O(n \times |N|) + O\left(\underbrace{n^2}_{\substack{\text{u.b. on } i, j \\ \text{as root}}} \times \underbrace{|N|}_{\substack{X \in N \\ \text{as root}}} \times \underbrace{|R| \times n}_{\substack{\text{in max} \\ \text{part} \\ \text{SE} \\ \{i, \dots, j\} - 1 \\ \text{part}}}}\right)$$

$$= O(n^3 |N| |R|)$$

More DP on PCFGs!!

Did given $x_1, \dots, x_n$, max $p(t)$ of tree over $x_1, \dots, x_n$

What if you want not max but total prob of all trees w/ $x_1, \dots, x_n$ leaves?

= prob of sent $x_1, \dots, x_n$

(every PCFG has a language model hiding inside of it)

Recall Viterbi: max prob of tag seq

Forward algo: prob of sentence = $\sum$ prob of all possible tag seqs

PCFGs CKY: max prob tree

Inside algo: prob of sentence = $\sum$ prob of all possible trees

$$\pi(i, j, X) = \sum_{t \in \mathcal{T}(i, j, X)} p(t)$$

max swapped for $\sum$
like Viterbi $\rightarrow$ Forward
in HMMs

i.e. is total prob of leaves $x_1, \dots, x_j$ given they're generated from an X at root (w/ any derivation)

$$\pi(i, j, X) = \sum_{t \in T(i, j, X)} p(t)$$

FP DP 1 $p(s) = \sum_{t \in T_G(s)} p(t) = \sum_{t \in T(1, n, S)} p(t) = \pi(1, n, S)$

FP DP 2 base case $\pi(i, i, X) = \begin{cases} q(X \rightarrow x_i) & \text{if } X \rightarrow x_i \in R \\ 0 & \text{otherwise} \end{cases}$

(max = sum when it's over 1 element)

Recursion

(smaller gaps $j-i$ done)

$$\pi(i, j, X) = \sum_{\substack{X \rightarrow YZ \in R \\ s \in \{i, \dots, j-1\}}} q(X \rightarrow YZ) \times \pi(i, s, Y) \times \pi(s+1, j, Z)$$

swap max $\rightarrow \sum$

In class problems

Q1 CFG not PCFG

want # of valid parse trees

table $\pi(i, j, X) = \# \text{ parse trees } \overset{\text{over}}{\text{subseq}} X_i, \dots, X_j \text{ with root } X$

FP DP 1 yes total # = $\pi(1, n, S)$

FP DP 2 recursion

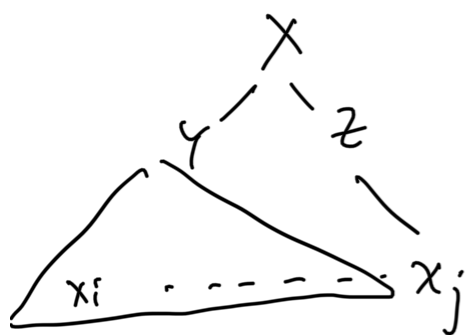
$$\pi(i, j, X) = \sum_{\substack{X \rightarrow YZ \in R \\ s \in \{i, \dots, j-1\}}} (\pi(i, s, Y) + \pi(s+1, j, Z))$$

Q2 left-branching trees

$\pi(i, j, X)$ almost same as CKY
 = max prob over left-branching trees
 of $p(t)$ that generate $x_i, \dots, x_j$
 & have root X

FP DP 2

$$\pi(i, j, X) = \max_{\substack{X \rightarrow YZ \\ \in R}} q(X \rightarrow YZ) \times \pi(i, j-1, Y) \times \pi(j, j, Z)$$



run time is $O(n^2 |N| |R|)$